Koushik Samanta

Senior Full-Stack Developer

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Professional Summary

- Full-stack developer with 9+ years at Codesigma Technology Services, building web, mobile and desktop products for clients in retail, healthcare, e-commerce, B2B distribution, media, gaming and education.
- Lead a team of 4 developers for the last 2.5 years: break client requirements into tasks, review code, guide junior developers and manage releases.
- Strong in React, Next.js, TypeScript and React Native on the front end, and Node.js, Express.js, NestJS, PHP and Laravel on the back end.
- Built multi-tenant SaaS products with role-based access control (RBAC), subscription billing and offline-first data, shipped to web, Android, iOS, Windows and macOS from shared codebases using Electron, Capacitor and Cordova.
- Hands-on with AWS (Amplify, Cognito, S3, Lambda, SES, CloudWatch), Firebase (Auth, Firestore, Cloud Functions, FCM), MongoDB, MySQL and Docker.
- Built headless CMS solutions on Payload CMS, along with custom WordPress themes and plugins, WooCommerce, Drupal and PrestaShop sites.
- Integrated payment gateways (Stripe, Razorpay, PayPal), OpenAI APIs, Azure AD SSO, Discord, Twitter, Google Maps and Mapbox APIs.
- Use AI tools in daily development: Claude Code for coding, refactoring, tests and deploy scripts, and Claude Design for UI mockups. Built an MCP integration that connects Claude with Payload CMS.
- Focus on page speed, accessibility and clean code: WCAG AA compliance, mobile Lighthouse scores of 80 to 90, and SQL query and index tuning.

Technical Skills

Languages	JavaScript (ES6+), TypeScript, PHP, SQL, HTML5, CSS3
Front-End	React, Next.js, Vite, React Native, Angular, React Query, jQuery, Bootstrap, Chart.js, HTML5 Canvas, AJAX
Back-End	Node.js, Express.js, NestJS, Laravel, CodeIgniter, REST APIs, Webhooks, Cron Jobs
Databases	MongoDB, MySQL, MariaDB, Cloud Firestore
Cloud & DevOps	AWS (Amplify, Cognito, S3, Lambda, SES, CloudWatch), Firebase (Authentication, Firestore, Cloud Functions, Hosting, FCM), Docker, Cloudflare, Git, CI/CD deployment pipelines
Cross-Platform	Electron.js (Windows, macOS), Capacitor, Apache Cordova, React Native (Android, iOS)
CMS & E-commerce	Payload CMS (headless), WordPress, WordPress Multisite, WooCommerce, Drupal, PrestaShop, Fork CMS
Integrations	Stripe, Razorpay, PayPal, OpenAI API, MCP, OAuth 2.0, Azure AD SSO, Google Maps & Places API, Mapbox, Discord API, Twitter API, OneSignal, FFmpeg, PHPMailer
Architecture	Multi-tenant SaaS, RBAC, MVC, Serverless, Headless CMS, Offline-first apps
AI Tools	Claude Code, Claude Design, OpenAI (ChatGPT) API

Professional Experience

Jan 2017 – Present **Codesigma Technology Services**

Jan 2022 – Present **Senior Software Developer**

- Lead a team of 4 developers since 2024: plan tasks from client requirements, review code and manage production releases.
- Develop and maintain SalePointPRO, a multi-tenant retail SaaS platform running on web, mobile and desktop.
- Built the Sivananda Ashram website on Payload CMS and now rebuilding its course booking platform with Next.js and NestJS.
- Deliver full-stack apps with React, Next.js, React Native, Node.js, AWS and Firebase, and client websites on WordPress, Drupal and PrestaShop.

Jan 2020 – Dec 2021 **Software Developer**

- Built inventory, employee and reporting modules in Laravel for TradeRunner as part of a shared team codebase.

- Wrote custom WordPress and WooCommerce plugins for AWS S3 uploads with PDF generation, SMTP email logging and live T-shirt customization.

Jan 2017 – Dec 2019 **Junior Software Developer**

- Built PHP applications in CodeIgniter and Laravel: a scripture search portal, billing, inventory and movie ticket booking systems.
- Converted HTML designs into custom WordPress and Fork CMS themes and added booking, store locator and media processing features.

Key Projects

Jan 2022 – Present **SalePointPRO – Multi-Tenant Retail Sales & POS Management Platform**

SaaS platform for retail businesses with separate Admin, POS, Customer, Employee, CRM, Kicker Point and Subscriber Control apps, available on web, Android, iOS, Windows and macOS.

Tech Stack: React, Vite, JavaScript (ES6), AWS Amplify, AWS Cognito, AWS S3, AWS Lambda, AWS SES, AWS CloudWatch, Firebase, Cloud Firestore, FCM, OneSignal, Electron.js, Apache Cordova, REST APIs, PayPal, OpenAI API

- Moved the platform from Firebase to React (Vite) and AWS Amplify, adding offline-first support and a structure that scales better and is easier to maintain.
- Built a multi-tenant subscription model priced by number of stores and warehouses, with PayPal for subscription payments.
- Added an AI FAQ assistant inside the Admin and POS apps using the OpenAI (ChatGPT) API with a custom chat window, so users get product help without leaving the app.
- Built AI-based expense entry: users upload an invoice or take a photo, and the app fills in vendor details, line items, taxes and totals automatically.
- Developed POS features for sales, refunds, exchanges, layaway sales, invoice and receipt printing, QR codes and QR-based customer feedback.
- Built inventory, warehouse, stock, customer, employee, expense, appointment, attendance, payroll and reporting modules.
- Set up role-based access with module-level permissions for store owners, employees, customers and system admins.
- Packaged desktop apps for Windows and macOS with Electron, and mobile apps for Android and iOS with Cordova.
- Use AWS Cognito for sign-in across all apps, S3 for files, Lambda (earlier Firebase Cloud Functions) for background jobs, SES, FCM and OneSignal for notifications, and CloudWatch for logging.
- Set up Amplify Hosting with custom domains and deployment pipelines for the production web apps.
- Work with product owners and QA engineers on new features, releases and production support across all platforms.

Jun 2025 – Present **Sivananda Ashram Neyyardam – Website & Course Booking Platform**

Official website for a yoga ashram with all content managed through a headless CMS, plus a new course booking platform in active development.

Tech Stack: React, Vite, Next.js, NestJS, Payload CMS, MongoDB, REST APIs, MCP, OAuth 2.0

AI Tools: Claude Code (AI-assisted coding), Claude Design (AI-assisted UI/UX design)

- Built and launched the website in React (Vite) with Payload CMS as the headless CMS, connected through REST APIs.
- Rebuilding the course booking platform with a Next.js front end and NestJS APIs, designed to scale and to add learning management features later.
- Built an MCP integration for Payload CMS: Claude saves SEO-ready blog drafts to the admin panel through OAuth 2.0 sign-in, and an editor reviews and publishes them.
- Built a course enquiry system that assigns staff automatically by course, sends email alerts, tracks contacts and follow-ups, and exports to Excel.
- Added a custom role-based access layer that controls view, add, edit, delete and export rights for each part of the CMS.
- Designed global schemas for shared layout sections and collections for Blogs, Courses and Programmes with image and video support.
- Tuned page speed, SEO and responsive layouts for desktop, tablet and mobile, and kept new booking flows in line with the ashram's existing processes.

Sep 2026 – Oct 2026 **TracSpare B2B Catalog – Request-a-Quote Spare-Parts Platform**

Price-free B2B catalogue for a tractor spare-parts exporter with more than 25,000 SKUs.

Tech Stack: PHP 8.3, WordPress, WooCommerce, custom plugin and theme, MariaDB, Docker, Cloudflare, JSON-LD

AI Tools: Claude Code (AI-assisted coding), Claude Design (AI-assisted UI/UX design)

- Built a quote-list flow on WooCommerce: visitors add parts and send an enquiry by email or WhatsApp, with reference numbers, minimum-order checks and a bulk part-list reader.

- Wrote custom part-number search with alternate numbers, a tractor finder and faceted filters across 25,000+ SKUs.
- Added background CSV product import with a browser-based uploader.
- Redesigned the site to WCAG AA and raised mobile Lighthouse scores to 80–90 with responsive images, lazy loading, a custom SVG icon set, Cloudflare cache purging and JSON-LD structured data.
- Set up Docker environments and push-to-deploy staging.
- Used Claude Code for implementation, refactoring, tests, deploy scripts and audits, checking every change against measured results; mocked up pages in Claude Design before building them.

Aug 2026 **Arogya – Multi-Platform Healthcare Management System**

Healthcare SaaS covering clinic, pathology lab and pharmacy operations, built for web, desktop and mobile from one codebase.

Tech Stack: Next.js, TypeScript, Electron, Capacitor, Firebase

AI Tools: Claude Code (AI-assisted coding), Claude Design (AI-assisted UI/UX design)

- Built a single Next.js codebase packaged as a web app, an Electron desktop app and a Capacitor mobile app.
- Designed a tenant-aware SaaS structure with role-based permissions for healthcare organisations and their staff.
- Built patient registration, profiles, medical history and document management.
- Developed appointment scheduling, doctor profiles and availability management.
- Added laboratory workflows for test orders, sample tracking and test results.
- Built pharmacy inventory with a medicine catalogue and stock monitoring.
- Developed billing with invoice generation, payment tracking and financial records.
- Used Firebase for authentication, data, file storage and backend processing.

Jul 2026 **Supply Hub – B2B Distributor Inventory & Order Management App**

Multi-tenant mobile app for distributors to manage stock, orders, deliveries and billing with their retailers.

Tech Stack: React Native, TypeScript, Firebase Authentication, Cloud Firestore, Firebase Cloud Storage, Cloud Functions, React Query

AI Tools: Claude Code (AI-assisted coding), Claude Design (AI-assisted UI/UX design)

- Built a multi-tenant data model that keeps each distributor's data separate.
- Developed the full order flow: product selection, checkout, status tracking and order history.
- Built inventory with batch management, low-stock alerts and stock movement history.
- Added billing with invoices, payment tracking, credit notes and customer ledgers.
- Built returns that update stock and financial records together.
- Added shipment and delivery tracking with separate roles for distributors, sales executives and delivery executives.
- Built dashboards with KPIs, charts, notifications and PDF/CSV export.
- Used React Query for server state and Cloud Functions for backend logic.

Jun 2026 **RidersMate – Group Motorcycle Ride Planning & Live Tracking App (Personal Project)**

Android app built for my friends' riding group to plan rides, track every rider live and split trip expenses.

Tech Stack: React, Vite, Capacitor, Firebase Firestore, Firebase Cloud Messaging (FCM), Mapbox APIs, Google Maps APIs, JavaScript (ES6), Text-to-Speech, Chart.js, Cloudflare

AI Tools: Claude Code (AI-assisted coding), Claude Design (AI-assisted UI/UX design)

- Built a live group map showing the route, stops and each rider's position, with GPS updates sent to Firestore every 60 seconds through background location.
- Added Fuel, Food, Rest and SOS alerts sent as FCM push notifications through a Cloudflare relay and read aloud with text-to-speech.
- Built ride planning for one-way, round-trip and multi-day rides with day-wise stops; Mapbox Directions API calculates distances and the ride timeline.
- Wrote a settlement algorithm that brings group expense payments down to the fewest possible transactions.
- Added group joining with 6-digit invite codes and rider roles: Lead, Marshal, Member, Pillion and Tail.
- Built expense splitting with categories, rider-wise bar charts and category-wise pie charts.
- Let riders switch between Mapbox and Google Maps; added place autocomplete, rider distance on the map and ride history.

Mar 2023 – Jun 2023 **eGameZone – Online Esports Tournament Platform**

Tournament platform where PUBG and Free Fire players join paid matches and receive prize money in their wallets.

Tech Stack: Node.js, Express.js, MongoDB, JavaScript, Razorpay, Discord API, Cron Jobs, Tidio Chat, ShareThis

- Built player wallets with Razorpay top-ups, withdrawals, balance tracking and transaction history.
- Automated result processing: the prize split is calculated and winnings are credited to player wallets when results are published.
- Built game, tournament and prize pool management for admins, and tournament booking paid from wallet balance.
- Integrated the Discord API to create a channel for each tournament, with cron jobs posting match details and updates.
- Set up role-based access with employee management and permission-based modules.
- Built player sign-up, profiles, tournament listing and detail pages, bookings and results.
- Designed MongoDB collections and queries for tournament and wallet data; added Tidio live chat and ShareThis.

Jul 2020 – Sep 2022 **TradeRunner – Business & Inventory Management Platform (Collaborative Project)**

Business management platform built by a shared development team; I owned several core modules and the company website.

Tech Stack: Laravel, PHP, MySQL, WordPress, WPBakery Page Builder, HTML5, CSS3, JavaScript

- Built the Inventory Management module in Laravel for products and stock.
- Developed the Employee Management module with employee profiles and admin features.
- Built business and inventory reports.
- Created the corporate website in WordPress with a custom theme and WPBakery, based on approved UI designs.
- Integrated my modules into the main application with the team and supported production releases.

Sep 2018 – Mar 2019 **UK Jewish Film Festival – Movie Ticket Booking & Festival Management System**

Online ticketing and festival management system for a UK film festival running screenings across several theatres.

Tech Stack: Laravel, PHP, MySQL, JavaScript, jQuery, Bootstrap, HTML5, CSS3, Stripe

- Built online ticket booking with real-time seat selection, Stripe payments and downloadable e-tickets.
- Built seat layout management so theatre staff can set availability by day, screen and seat.
- Added booking administration with payment history, booking status and refund processing.
- Developed movie management with trailers, posters, genres, categories and screening schedules.
- Set up role-based access for admins, employees and theatre-specific staff.
- Added movie search and filters by theatre, genre, category and screening date.
- Built customer accounts with profiles and booking history, and tuned Eloquent queries for booking traffic.

Apr 2018 – Aug 2018 **DRUK – Inventory & Sales Management System**

Enterprise system for managing products, stock, customers and sales orders.

Tech Stack: Laravel, PHP, MySQL, HTML5, CSS3, Bootstrap, JavaScript, jQuery

- Built inventory management with a product catalogue, stock levels and stock tracking.
- Developed sales order management to create, update and manage customer orders.
- Built sales summary, inventory status and operations reports.
- Set up role-based access with configurable roles and module permissions, plus employee and customer management.
- Designed a normalised MySQL schema with Eloquent ORM and tuned queries for large inventory and sales data.
- Gathered requirements with stakeholders and delivered new features and production support.

Jan 2017 – Feb 2018 **SearchGurbani – Gurbani Search & Digital Scripture Portal**

Search portal for Sikh scripture with multilingual text and translations.

Tech Stack: PHP, CodeIgniter, MySQL, HTML5, CSS3, JavaScript, jQuery, Bootstrap, AJAX, Git

- Built keyword search with filters to find verses quickly across multilingual content.
- Designed the MySQL schema for scripture text, translations and metadata.
- Tuned SQL queries and indexes for faster search results.
- Built admin modules to manage scripture content and translations.
- Added AJAX-based search and page updates, with responsive pages in Bootstrap and jQuery.
- Wrote reusable CodeIgniter MVC modules with form validation, sessions and authentication.

Other Projects

Sep 2023 **Event Management Company – Portfolio & Booking Website**

Tech Stack: Firebase Hosting, Cloud Firestore, Firebase Modular Web SDK, HTML5, CSS3, JavaScript

- Built a single-page portfolio site on Firebase Hosting with a custom domain and SSL.
- Event enquiries are stored in Firestore, and the Firestore Send Email extension notifies the events team on each booking.

Aug 2023 **Dumi Homestay – Portfolio & Booking Website**

Tech Stack: Firebase Hosting, Cloud Firestore, Firebase Modular Web SDK, HTML5, CSS3, JavaScript

- Built a single-page homestay website with smooth-scroll navigation, hosted on Firebase with a custom domain and SSL.
- Booking requests are saved to Firestore and emailed to the owner through the Send Email extension.

Jul 2023 **EaseCommerce CRM – Customer Relationship Management Platform (Collaborative Project)**

Tech Stack: Next.js, NestJS, MongoDB, TypeScript, REST APIs, Webhooks

- Built webhook integrations that bring customer messages from Facebook, Instagram and Google into one CRM inbox.
- Added chat assignment to support staff, text and file replies, and chat status and resolution.
- Developed the Employee Management, Task Management and Sales Report modules.

Feb 2023 **Lalic – E-commerce Website**

Tech Stack: PrestaShop, PHP, MySQL, HTML5, CSS3, JavaScript, jQuery, Bootstrap

- Built a custom PrestaShop theme from HTML designs for product listing, product detail, cart and checkout pages.
- Configured the checkout flow and shipping modules to match the client's delivery rules.

Jan 2023 **Sunrise Solar Power – Solar Energy Solutions Website**

Tech Stack: WordPress, PHP, MySQL, JavaScript, jQuery, Bootstrap

- Built a solar cost calculator that estimates installation cost and yearly savings against grid electricity in real time.
- Converted HTML designs into a custom responsive WordPress theme.

Dec 2022 **TaxSigma – Tax & Financial Services Website**

Tech Stack: WordPress, PHP, MySQL, JavaScript, jQuery, Bootstrap

- Built a custom WordPress theme with Home, Services, Careers, Blog and Contact pages, all editable from the admin panel.
- Improved page load speed and set up SEO-friendly URLs.

Nov 2022 **Doggies & Co. – Pet Boarding & Day Care Booking Website**

Tech Stack: Drupal, PHP, MySQL, HTML5, CSS3, JavaScript, jQuery

- Built a custom Drupal booking module for pet boarding and day care, with a backend view of all bookings.
- Added email confirmations to customers and admins, and converted HTML designs into a custom Drupal theme.

Oct 2022 **Omexom – Corporate Gamification & Quiz Platform (Collaborative Project)**

Tech Stack: Angular, TypeScript, HTML5, CSS3, Bootstrap, Azure Active Directory

- Added Azure AD single sign-on so only employees with company email accounts can sign in.
- Built the results module in Angular showing quiz scores, rankings and player performance.

May 2020 – Jun 2020 **T-Shirt Customization Plugin (WooCommerce)**

Tech Stack: WordPress, WooCommerce, PHP, MySQL, JavaScript, jQuery, AJAX, HTML5 Canvas

- Built a WooCommerce plugin where customers choose country combinations and see a live T-shirt preview drawn with HTML5 Canvas.
- Hooked into Add-to-Cart to save the final design image through cart, checkout and order.
- Added admin settings to map countries to front, back, left and right placements for each product.

Apr 2020 **SMTP Email Management & Delivery Plugin**

Tech Stack: WordPress, PHP, MySQL, PHPMailer, JavaScript, jQuery

- Built a WordPress SMTP plugin on PHPMailer with admin settings for server, authentication, encryption and sender details.
- Added an email log with delivery status tracking and one-click resend.

Mar 2020 **AWS S3 File Upload & PDF Automation Plugin**

Tech Stack: WordPress, PHP, MySQL, AWS S3 SDK, JavaScript, jQuery

- Built a WordPress plugin that uploads form files to Amazon S3, creates a PDF of each submission and emails it to the customer and staff.
- Added admin settings for S3 credentials and a screen to view submissions with their files.

Feb 2020 **JantorMantor – Bengali News & Blogging Portal**

Tech Stack: WordPress, PHP, MySQL, JavaScript, jQuery, Bootstrap

- Built a Bengali news portal on a custom WordPress theme with category pages and infinite scroll on articles.
- Added configurable ad slots on category and article pages.

Dec 2019 – Jan 2020 **Bridal Dress Portfolio Website (Multi-Brand Portfolio)**

Tech Stack: Fork CMS, PHP, MySQL, JavaScript, jQuery, Bootstrap

- Built one Fork CMS website for two bridal brands, with custom themes and portfolio modules for dress collections.
- Matched the approved design pixel for pixel on desktop, tablet and mobile.

Nov 2019 **DenverPhoneDoctor – Mobile Repair Service Booking Platform**

Tech Stack: WordPress, PHP, MySQL, JavaScript, jQuery, Google Maps Geocoding API, Google Places API

- Built a store locator on Google Maps that finds repair centres within a chosen radius such as 50 or 100 km.
- Added appointment booking where customers pick a nearby store and service, with Places address autocomplete and store coordinates filled in through the Geocoding API.

May 2019 – Oct 2019 **Radio & Video Streaming Platform (WordPress Multisite)**

Tech Stack: WordPress Multisite, PHP, MySQL, FFmpeg, JavaScript, jQuery, Cron Jobs, Twitter API

- Used FFmpeg to read media metadata, create thumbnails and generate missing video resolutions (360p to 1080p) from an admin dashboard.
- Wrote a cron job that tweets the song currently playing, with artwork and stream link, through the Twitter API.
- Added new features and performance improvements on the live multisite network.

Apr 2019 **JanitorUtah – Cleaning Services Website**

Tech Stack: WordPress, PHP, MySQL, JavaScript, jQuery, Bootstrap

- Converted a static HTML site into a custom WordPress theme for a cleaning services company.
- Built a booking system with Custom Post Types and an admin screen to manage requests.

Mar 2018 **Cable Operator Billing & Customer Management System**

Tech Stack: Laravel, PHP, MySQL, Bootstrap, JavaScript

- Built monthly billing with customer-wise payment tracking, due and paid reports, and monthly and yearly sales reports.
- Added role-based access with configurable permissions and subscriber management.

Training

2016 **Software Development Training (3 months)**, *Codesigma Technology Services*

Education

2012 – 2015 **Bachelor of Computer Applications (BCA)**, *Vidyasagar University, Midnapore.* Score: 57.93%

2012 **Higher Secondary (WBCHSE)**, *Bagnan High School, Howrah.* Score: 53.40%

2010 **Secondary (WBBSE)**, *Bagnan High School, Howrah.* Score: 60%

Personal Details

Languages English, Bengali, Hindi

Hobbies Trekking, Travel, Bike Riding